

HENRIQUE SCHECHTER VERA

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EDUCATION

Princeton University B.S.E. in Computer Science

Sep. 2019–May 2023

Honors: Summa Cum Laude, Phi Beta Kappa (early inductee), Tau Beta Pi (early inductee)

Minors in Applied & Computational Mathematics, Statistics & Machine Learning, Optimization & Quantitative Decision Science

Overall GPA: 3.99, Departmental GPA: 4.0

- **Graduate coursework:** Advanced Algorithm Design, Data Structures and Graph Algorithms
- **Undergraduate:** Natural Language Processing (NLP), Machine Learning Theory, Machine Translation, Optimization
- **Natural languages:** (native/fluent proficiency) English, Spanish, Portuguese, Guarani
- **Programming languages:** Java, Python, C, C++, Go, SQL, HTML, CSS, JavaScript, Assembly, R, MATLAB

RESEARCH

EmbeddingGemma: Powerful and Lightweight Text Representations

2025

Google Deepmind ([Paper](#))

We introduce a state-of-the-art (SOTA) 300M parameter, open text embedding model. Our innovative training recipe includes encoder-decoder initialization, geometric embedding distillation, a spread-out regularizer, and model merging with Bayes-optimized mixtures. Evaluated on the Massive Text Embedding Benchmark (MTEB) across multilingual, English, and code domains, we achieve SOTA results, outperforming prior top models with fewer than 500M parameters and providing performance comparable to models double the size. This lead persists when quantizing model weights or truncating embedding outputs.

Gemini Embedding: Generalizable Embeddings from Gemini

2024–2025

Google Deepmind ([Paper](#))

We introduce a text embedding model with capabilities spanning numerous languages and modalities. We employ Gemini for several data curation steps: filtering low-quality examples, determining relevant positive and negative passages for retrieval, and generating rich synthetic datasets. Evaluated on the Massive Text Embedding Benchmark (MTEB), which includes > 100 tasks across 250+ languages, we substantially outperform prior SOTA models and surpass specialized domain-specific models.

Structured Pruning for Pre-Trained Models

2022–2023

Adviser(s): Prof. Danqi Chen, Mengzhou Xia (Princeton Language and Intelligence) (github.com/princeton-nlp/CoFiPruning)

We adapt and evaluate a novel structured pruning method on pre-trained models. The key insight is to jointly prune coarse-grained (e.g. layers) and fine-grained modules (e.g. hidden units), controlling the pruning decision of each parameter with masks of different granularity. We also devise a dynamic, layerwise distillation strategy for use during pruning.

Selling to a Sophisticated No-Regret Buyer

2022–2023

Adviser(s): Prof. Matt Weinberg, Second Reader: Prof. Mark Braverman ([Paper](#))

Consider a repeated single item auction with a single buyer who bids according to a no-regret online learning algorithm. Say the buyer cleverly applies recency bias or k -switching to their algorithm. In both settings, we (1) find for what parameter values the buyer remains no-regret, and (2) provide an auction which provably yields near-maximal revenue.

Extending Randomized Single Elimination Bracket to Multiple Prize Vectors

2022–2023

Adviser(s): Prof. Matt Weinberg, Prof. Huacheng Yu ([Paper](#))

We extend Randomized Single Elimination Bracket (RSEB) [Schneider et al., 2017] to tournaments with multiple prizes, creating two new tournament ranking rules: *Randomized Complete Bracket* and *Randomized Recursive Bracket*. We prove various guarantees for fairness and manipulability under these rules.

Exploring the Role of Contextualization in Dynamic Contextualized Word Embeddings (DCWEs)

2022

Adviser(s): Prof. Karthik Narasimhan ([Paper](#))

DCWEs incorporate both linguistic and extralinguistic context, combining a contextual language model with a dynamic model of time and social space. We evaluate the effect of the dynamic component on various contextual and non-contextual models on a sentiment analysis task, finding that it improves performance on non-contextual models, but not (most) contextual ones.

Cannabis Analysis: Understanding Strain Names through NLP

2021–2022

Adviser(s): Prof. Christiane Fellbaum, Dr. Jérémie Lumbroso ([Paper](#))

We create and publish the largest public [cannabis strain dataset](#). We analyze patterns in strain names strains' ratings, effects, flavors, and genealogy. We find that popular strains do not have distinct lexical distributions or sentiment values, strains' parents account for their popularity, and strains named like their parents tend to differ in popularity from their parents.

Lyric-Based Automatic Playlist Continuation Using Feature Engineering

2021

Adviser(s): Prof. Srinivas Bangalore ([Paper](#))

We develop a novel automatic playlist continuation system. We establish a method of featurizing songs based on vocabulary, structure, orientation, style, and semantics, and validate it using dimensionality reduction. We then train a neural network to determine song compatibility using 1M Spotify playlists, yielding a 4x improvement in recommendations against the benchmark.

EXPERIENCE

- Google DeepMind, Senior Research Engineer** *November 2025–Present*
- Conducting fundamental research in multimodal representation learning, computer vision, and language understanding
- Google, Senior Machine Learning Engineer** *July 2023–November 2025*
- Drove research in natural language understanding and landed state-of-the-art innovations across Alphabet products
 - Worked on the research, development, and deployment for Google Search’s query understanding models (XXXXB searches/day)
- Google, Software Engineering Intern** *May 2022–August 2022*
- Completely re-implemented team’s flagship neural semantic parsing model
 - Designed architecture alongside researchers, implemented the model, and built training and evaluation infrastructure
 - Led initiative and communication with research teams to open-source our model’s classifier and new debugging tools
- Google, Software Engineering Intern** *May 2021–August 2021*
- Developed end-to-end features for electric vehicles in Google Maps, using NLP to create voice queries for Google Assistant
 - Served as direct liaison between two teams involved, driving communications with UX, PM, and SWE correspondents
 - Went beyond assigned tasks: Successfully lead work with a fellow intern to jumpstart a related project
 - Fixed a year-old bug directly affecting user experience. Answered several intern questions on Q&A site and groups
- Landed (Y Combinator W19), Software Engineering and Product Management Intern** *June–Aug. 2020*
- Created and shipped standalone, customer-facing pricing tool using JavaFX and CSS. Determines cost based on amount, location, urgency, and job of the hires. Continues to be used in meetings with prospective clients
 - Led team through product development life cycle for various projects, from ideation to deployment to analysis
 - Introduced successful features to platform with 10k+ users, such as a preference-based jobs page and filming overlay tips
 - Created data analytics dashboards and tracked key metrics to identify growth opportunities, evaluate product performance, and deliver concrete insights. Used SQL (PostgreSQL), Excel, and Google Data Studio
- Princeton Computer Science, Research TA, Head TA, Instructor, Grader, Tutor, Interactor** *Feb. 2020–2023*
- Analysis of Succinct Data Structures (Research TA), Machine Translation (Head TA), Machine Learning (TA + grader), Reasoning About Computation (Tutor), Algorithms and Data Structures (TA + grader), Programming Systems (TA), Introduction to Computer Science (Instructor + TA + Grader)
 - Research TA: help students with CS research projects, give feedback for proposals and papers (with Prof. Huacheng Yu)
 - Head TA: Created and led part of curriculum for intro NLP course, held office hours, monitored discussion forum
 - Instructor: Led an introductory CS class twice a week, teaching 15 Princeton undergraduates
Evaluated applications for future Undergraduate Course Assistants
 - Interactor: Worked with a professor as academic advisors for 1st-year CS majors
- Po Paraguay, Product and UX Intern** *June 2017*
- Conducted user interviews and tested prosthetic performance to refine product designs
 - Created public product pages for our 3D-printed prosthetics, with multilingual assembly guides and videos of prosthetic functionality, increasing reach and accessibility. Recruited and managed 60 volunteers for 2 years afterward

ACTIVITIES

- Princeton Data Science, Officer** *2021–2022*
- Coordinated educational student events and speaker talks with experts, and established formal partnerships with companies
- Princeton Puzzle Hunt, Web Developer** *2020–2022*
- Learned HTML, CSS, and JavaScript to help reconstruct and manage the entire website for this fully digital event
- Jane Street ETC; Facebook; HackPrinceton; ReHack.co, Competitor; Fellow; Coder; Team Captain** *2019–2021*
- Created an algorithmic trading bot. Learned finance and socket programming. Placed 6th out of 22 team
 - Learned advanced CS over 10 weeks with FB Software Engineers
 - Built privacy app that utilizes ML-based facial recognition to protect one’s computer from unwanted users
- Tigers For Nassau, Global Research and Consulting Group, Technology Strategist and Consultant** *2020–2021*
- Created digital solutions (delivery platforms, ordering systems) to adapt businesses to the COVID-19 crisis
- TechTogether New York, Organizer** *2020*
- Coordinated and carried out corporate deals and partnerships. Designed our sponsorship package
- Student Government and Student Council, Class Representative, Class Officer (Treasurer)** *2016–2019*
- Supervised marketing, partnerships, finances, staffing, and logistics for events with 40,000+ attendees and 1,500+ athletes

AWARDS

- Phi Beta Kappa** Early Inductee, Sep. 2022
- Tau Beta Pi** Early Inductee, Sep. 2021
- Paraguay National Math Olympiad** (2x) Silver Medal and Departmental Champion (2017–2018)
- International Math Kangaroo** 1st, 3rd, 3rd, 5th, 5th, and 7th place nationally (2010–2019)